Date.....

Expt. No..... Page No.....

Class Test - 3.

Q1) Here: ~~V~~ $L = T - V$

$$\therefore V = \frac{1}{2} mgl(\theta_2^2 + \theta_1^2) + \frac{1}{8} kl^2(\theta_2 - \theta_1)^2$$

$$= \frac{1}{2} mgl(\theta_1^2 + \theta_2^2) + \frac{1}{8} kl^2(\theta_1^2 + \theta_2^2 - 2\theta_1\theta_2)$$

$$= \frac{1}{2}\left[\theta_1^2\left(mgl + \frac{kl^2}{4}\right) + \theta_2^2\left(mgl + \frac{kl^2}{4}\right) - \theta_1\theta_2\left(\frac{1}{2}kl^2\right)\right]$$

$$\therefore V = \begin{pmatrix} mgl + \frac{kl^2}{4} & -\frac{kl^2}{4} \\ -\frac{kl^2}{4} & mgl + \frac{kl^2}{4} \end{pmatrix}$$

$$T = \frac{1}{2} ml^2(\dot{\theta}_1^2 + \dot{\theta}_2^2)$$

$$\therefore M = \begin{pmatrix} ml^2 & 0 \\ 0 & ml^2 \end{pmatrix}$$

Let ω^2 be the normal modes of oscillation

$$\therefore |V - \omega^2 M| = 0$$

$$\Rightarrow \begin{vmatrix} mgl + \frac{kl^2}{4} - \omega^2 ml^2 & -\frac{kl^2}{4} \\ -\frac{kl^2}{4} & mgl + \frac{kl^2}{4} - \omega^2 ml^2 \end{vmatrix} = 0$$

Teacher's Signature : _____

$$\Rightarrow \left(mgl + \frac{kl^2}{4} - \omega^2 ml^2\right)^2 = \left(\frac{kl^2}{4}\right)^2$$

case I:

(b) $\Rightarrow$ $$mgl + \frac{kl^2}{4} - \omega^2 ml^2 = \frac{kl^2}{4}.$$

$$\Rightarrow \quad mg\cancel{l} = \omega^2 m l^{\cancel{2}} \quad \Rightarrow \quad \boxed{\omega_1^2 = \frac{g}{l}}$$ (Ans)

case II: $$mgl + \frac{kl^2}{4} - \omega^2 ml^2 = -\frac{kl^2}{4}.$$

$$\Rightarrow \quad \omega^2 ml^2 - mg\cancel{l} = \frac{kl^2}{2}$$

$$\Rightarrow \quad \omega^2 = \frac{\frac{kl^2}{2} + mg}{ml}.$$

$$\boxed{\omega_2^2 = \frac{g}{l} + \frac{k}{2m}}$$ (Ans)

(ii) Let the amplitudes be $A_i = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$

$\omega_1^2 = \frac{g}{l}$:

$$(V - \omega_1^2 M) A_1 = 0.$$

$$\Rightarrow \begin{pmatrix} \frac{kl^2}{4} & -\frac{kl^2}{4} \\ -\frac{kl^2}{4} & \frac{kl^2}{4} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0.$$

$$\Rightarrow \quad \boxed{a_1 = a_2}$$

$$\therefore \quad \boxed{A_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}}$$ (Ans)

Date

Expt. No. Page No.

$$\omega_2^2 = \frac{g}{l} + \frac{k}{2m}$$

$$ml^2\omega_2^2 = ml^2\left(\frac{g}{l} + \frac{k}{2m}\right) = mgl + \frac{kl^2}{2}$$

$$\therefore (V - \omega_2^2 M)(A_2) = 0$$

$$\Rightarrow \begin{pmatrix} -\frac{kl^2}{4} & -\frac{kl^2}{4} \\ -\frac{kl^2}{4} & -\frac{kl^2}{4} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0 \quad \Rightarrow a_1 = -a_2$$

$$\boxed{A_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}}$$ (Ans)

(2) ~~$T = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2(\dot{x}^2 + l^2\dot{\theta}^2 + 2\dot{x}\dot{\theta}l) +$~~

(1)

$$M = \begin{pmatrix} m_1+m_2 & m_2 l \\ m_2 l & m_2 l^2 \end{pmatrix}$$

[Assuming $k = k_1$]

$$V = \begin{pmatrix} k & 0 \\ 0 & m_2 g l \end{pmatrix}$$

$$\therefore |V - \omega^2 M| = 0$$

$$\Rightarrow \begin{vmatrix} k - \omega^2(m_1+m_2) & -\omega^2 m_2 l \\ -\omega^2 m_2 l & \cancel{m_2 g l^2 - \omega^2} m_2 g l - \omega^2 m_2 l^2 \end{vmatrix} = 0.$$

Teacher's Signature :

$$\Rightarrow [k - \omega^2(m_1+m_2)][m_2 g l - \omega^2 m_2 l^2] = (\omega^2 m_2 l^2)^2$$

$$\Rightarrow [k - \omega^2(m_1+m_2)](g - \omega^2 l) = \omega^4 m_2 l$$

$$\Rightarrow gk - \omega^2 kl - \omega^2 g(m_1+m_2) + \omega^4 l(m_1+m_2) = \omega^4 m_2 l$$

$$\Rightarrow \omega^4 m_1 l - \omega^2[g(m_1+m_2) + kl] + gk = 0.$$

$$\therefore \omega^2 = \frac{g(m_1+m_2) + kl \pm \sqrt{[g(m_1+m_2)+kl]^2 - 4gk m_1 l}}{2m_1 l}$$

If $m_1 = m_2 = m$:

$$\omega^2 = \frac{2mg + kl \pm \sqrt{(2mg+kl)^2 - 4gklm}}{2ml}$$

$$= \left(\frac{g}{l} + \frac{k}{2m}\right) \pm \frac{\sqrt{4m^2g^2 + k^2l^2}}{2ml}$$

$$\omega_1^2 = \left(\frac{g}{l} + \frac{k}{2m}\right) + \sqrt{\frac{g^2}{l^2} + \frac{k^2}{4m^2}}$$

$$\omega_2^2 = \left(\frac{g}{l} + \frac{k}{2m}\right) - \sqrt{\frac{g^2}{l^2} + \frac{k^2}{4m^2}}$$

[Ans]

(ii) ω_1^2:

$$\boxed{p = \sqrt{\frac{g^2}{l^2} + \frac{k^2}{4m^2}}}$$

$$(V - \omega_1^2 M) A_1 = 0.$$

$$\Rightarrow \begin{pmatrix} \cancel{k} - \frac{2mg}{l} - \cancel{k} + 2mp & -mg - \frac{kl}{2} - ml\sqrt{\frac{g^2}{l^2} + \frac{k^2}{4m^2}} \\ -mg - \frac{kl}{2} - ml\sqrt{\frac{g^2}{l^2} + \frac{k^2}{4m^2}} & \cancel{mgl} - \cancel{mgl} - \frac{kl^2}{2} - ml^2\sqrt{\frac{g^2}{l^2} + \frac{k^2}{4m^2}} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0.$$

$$\Rightarrow \begin{pmatrix} \frac{2mg}{l} + 2mp & +mg + \frac{kl}{2} + mlp \\ mg + \frac{kl}{2} + mlp & \frac{kl^2}{2} + ml^2 p \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0.$$

Date.....

Expt. No..... Page No.....

$$\Rightarrow \left(\frac{2mg}{l} + 2mP\right) a_1 + \left(mg + \frac{kl}{2} + mlP\right) a_2 = 0$$

$$\Rightarrow \quad a_2 = -\frac{\left(\frac{2mg}{l} + 2mP\right)}{\left(mg + \frac{kl}{2} + mlP\right)} a_1$$

$$\therefore A_1 = \begin{pmatrix} 1 \\ -\dfrac{\left(\frac{2mg}{l} + 2mP\right)}{\left(mg + \frac{kl}{2} + mlP\right)} \end{pmatrix}$$ (Ans)

Similarly A_2 can be calculated.